

Self-expression property theory guided multi-modal brain graph learning

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ABSTRACT

Integrating multi-modal brain graphs for brain disorder analysis has attracted increasing attention recently. However, existing multi-modal brain graph learning methods suffer from two important problems: first, they randomly use simple connectivity profiles as attribute features of brain graphs, causing an insufficient description of brain connectivity networks. In addition, unobvious differences between healthy and patient brain graphs require to learn discriminative feature representations of single-modal brain graphs, which is rarely considered in previous methods. Second, mining the complex relationship between modalities is significant for achieving deep fusion, but most of the existing methods adopt a simple fusion way, like the attention mechanism, and cannot recognize important brain regions and connections associated with disorders for interpretable analysis. To address the problems above, in this work, we integrate cross-scale contrastive graph neural networks and self-expression property theory to learn multi-modal and multi-attribute brain graph representations for disorder analysis. Specifically, we extend multi-attribute features for characterizing brain graphs and design cross-scale contrastive graph neural networks to handle multi-attribute brain graphs to enhance discriminative multi-level representations for each modality. Then, we introduce the self-expression property theory to innovatively extract the complementary relationship of multi-modal brain graphs while generating the consensus graph to learn the common feature. The consensus graph is an interpretable brain graph to explore salient brain regions and connections related to disorders. We adaptively fuse the common feature and each modality feature using the attention mechanism to learn optimal representations of multi-modal brain graphs for disorder prediction. Extensive experiments on three disorder datasets verify the effectiveness of our method, especially pathogenic analysis results are in line with medical findings, which has the potential to promote brain disorder research.

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1. Introduction

With the development of neuroimaging technology, such as functional Magnetic Resonance Imaging (fMRI) and Diffusion tensor imaging (DTI), exploring brain connectivity networks has achieved great progress, further promoting brain disorder research. Brain connection networks (brain graphs) are commonly constructed from neuroimaging data by defining different brain regions of interest (ROIs) as nodes and connection relationships between ROIs as edges according to previous works [1,2]. Traditional brain network analysis methods rely on statistical evaluations [3] and shallow models [4] to measure brain connections for disorder analysis, like node degree, global efficiency, and Support Vector Machine (SVM). This not only cannot extract hidden structural patterns of brain graphs, but also entails the problem of poor performance for downstream tasks. Recently, graph neural networks (GNNs) [5], designed to process graph structural data and learn graph representations, have been widely applied to brain graph representation learning for disorder analysis.

Most of the brain graph learning methods focus on designing new graph convolution layers to learn better brain graph representations for disorder prediction [1,6], i.e., brain graph classifications, or extract disorder-specific ROIs and connections for pathogenic analysis [7,8]. Although these methods make great explorations for deep brain graph learning, they only focus on a single modality of neuroimaging, fMRI or DTI. Multi-modal brain graphs can provide more complementary information than a single modality for disorder analysis. However, existing methods aim to learn a common feature representation between the modalities [9–11] by a simple fusion strategy, such as the attention mechanism and tensor decomposition. Other methods [12] consider utilizing the feature information of one modality to guide the learning of another modality.

However, the following research problems are rarely discussed. First, brain graphs, unlike other commonly used graph datasets [5] that have natural node attribute features, are constructed from brain neuroimaging, and node attribute features are missing. To employ GNNs in brain graph representation learning, most existing methods [7,13,14] directly use structure information, e.g., the adjacency matrix, as attribute features of brain region nodes for the graph convolution. However, such simple attribute features may be insufficient for characterizing brain graph properties, leading to poor representation quality. In addition, there are small differences between healthy and patient brain graphs because healthy brain graphs have similar structures, and several abnormal brain regions or connections may cause patient brain graphs. Thus, brain graph representations learned in the above way are too similar to distinguish, which limits the performance of disorder prediction. Integrating various attribute features for brain graph description is a promising way to highlight the characteristics of brain graphs. This also results in learning discriminative brain graph representations. But how to effectively utilize and fuse multi-attribute features for brain graph learning is worth exploring.

Furthermore, some literatures [15,16] have shown that the relationship between brain graphs of different modalities is complex. For example, brain graphs between fMRI and DTI are heterogeneous at the region-level, and their feature representations are highly complementary. The simple fusion methods in the previous methods are suboptimal. They cannot deeply fuse multi-modal data to improve brain graph representations. Recent methods [13,17] propose using contrastive mutual fusion to address this problem, or RH-BrainFS [14] designs a transformer-based fusion bottleneck to capture the regional heterogeneity between structural and functional brain graphs. But they ignore the key problem mentioned that when the quality of single modality features cannot be guaranteed, the fusion result will be worse.

Recently, self-expression property theory has achieved great performance in the multi-view subspace clustering task [18,19]. They mainly utilize this theory to learn a coefficient matrix for capturing complementary information across views. This undoubtedly brings new insights for multi-modal brain graph learning. But there is a research gap. First, brain graph structures are complex, e.g., connectivity sparsity and high-order relationships, which influence the effectiveness of the learned coefficient matrix. Second, how to utilize the advantage of the coefficient matrix to recognize salient brain regions and connections for pathogenic analysis is left unexplored.

To address these problems above, in this work, we propose a self-expression property theory guided multi-modal brain graph learning framework for disorder prediction and pathogenic analysis (**BrainMM**). First, inspired by works [2,20] that explore the construction of node attribute features for non-attributed graphs, we extend multi-attribute features for brain graphs and design cross-scale contrastive graph neural networks to enhance multi-level representations of brain graphs for each modality. Specifically, the contrastive learning mechanism utilizes different attribute feature views to build the cross-scale contrast between node-level and graph-level representations. Thus, this not only helps to learn discriminative graph-level representations for the single modality but also benefits the complementary information mining of multi-modal brain graphs based on learned node-level representations.

Then, we introduce the self-expression property theory [21,22] to extract the complementary information of multi-modal brain graphs while generating the consensus brain graph based on node-level representations of all modalities. Besides, to guarantee the effectiveness of the consensus graph, we integrate the high-order structure information from each modality to promote the consensus graph generation. Furthermore, the feature representations of each modality are important, and different subjects are described by relying on the real quality of each part representation. We utilize the attention mechanism to adaptively fuse the common feature and each modality feature to learn optimal brain graph representations for disorder prediction. In addition, the consensus graph is an interpretable brain graph, and we design the unsupervised Class Activation Map (CAM) to recognize salient brain regions and connections for pathogenic analysis. We highlight the key contributions of BrainMM as follows:

- We design cross-scale contrastive graph neural networks based on multi-attribute features to enhance multi-level brain graph representations of each modality.

- To the best of our knowledge, this is the first work to introduce the self-expression property theory to extract the complementary information for multi-modal brain graphs. We are the first to utilize the consensus brain graph as the interpretable brain graph for pathogenic analysis.
- We adaptively fuse the common feature and each modality feature by the attention mechanism to learn optimal brain graph representations for disorder prediction.

2. Related works

2.1. GNNs based brain graph learning

Single Modality Based Brain Graph Learning. Traditional brain graph analysis methods [3] rely on graph theory to measure the condition of brain connections, such as node degree and clustering coefficient. Other shallow methods [24] utilize tensor factorization technology to obtain informative representations of brain graphs for classification tasks. In addition, the method [25] utilizes a convolutional neural network and a recurrent neural network to learn feature representations from brain networks, respectively, for Alzheimer's disease detection. For GNN-based methods, hi-GCN [26] combines graph convolutional networks (GCN) [27] and global pooling to capture single graph structures and the correlation among subjects for brain graph representations. BRAINNETTF [28] designs a brain network transformer model to capture the complex structure of brain networks. DGCL [29] proposes a new brain network construction paradigm to address brain region localization and brain network connection problems by combining graph diffusion and contrastive learning. FC-HGNN [30] first constructs an individual brain graph and a heterogeneous population graph, and then designs a new graph convolution layer to identify biomarkers for individual brain graphs. Then, a graph transformer is used to learn feature embedding for population graphs. Other works [31,32] focus on addressing the problem of data scarcity by introducing model pre-training or meta-learning methods. In addition, some works also propose a series of interpretable brain graph analysis models [6–8]. For example, IBGNN [7] proposes a globally shared explanation generator across the group to highlight salient ROIs and important connections. To capture the temporal dynamics of brain graphs, most of the existing methods [33] first construct dynamic brain graphs based on the division of time series for fMRI and design spatio-temporal graph convolution networks by combining LSTM and GNNs to learn brain graph representations. Specifically, the newest work [34] re-examines the effectiveness of graph deep learning for brain graph learning, concluding that simple message aggregation mechanism cannot improve predictive performance. In summary, these methods focus on brain graph analysis from a single neuroimaging modality.

Multi-modal Brain Graph Learning. Current multi-modal graph learning has been widely studied in many fields, such as incomplete multi-modal learning in conversation data [35], multi-modal graph fusion for freezing of gait detection [36], GCN-based multi-modal noise contamination for recommendation [37], and drug-drug interaction prediction [38]. For brain graph data, as shown in the comparison in Table 1, some multi-modal brain graph learning models aim to fuse multi-modal information to learn a common representation. For example, MGNet [9] first uses tensor representation technology to fuse multi-modal brain graphs, then inputs them into multiplex GCNs to learn common embeddings. Cross-GNN [17] learns a compositional space by constructing a dynamic multi-modal graph considering inter-modal relationships. The method [10] utilizes the cross-attention mechanism to extract complementary information between fMRI and DTI. The method [39] integrates fMRI, DTI, and sMRI into a brain graph and designs a mask GNN to learn feature representations for predictive analysis.

On the other hand, some methods directly combine different modality representations. BrainNN [13] utilizes contrastive learning between two modalities to enhance brain graph representations of each modality and combines them. Similarly, BMCL [23] designs two encoder-decoder architectures to learn region representations of each modality and constructs region-level contrast to enhance them. Then, all modality representations are combined by the concatenation operation. Other methods utilize one modality of information to guide another modality's learning to achieve multi-modal fusion. For example, SCP-GCN [12] defines DTI modality as brain graph structures and fMRI modality as node features. Then, it proposes a Siamese architecture to guide the multi-modal brain graph learning process for community structure mining. Differently, RH-BrainFS [14] designs a fusion bottleneck to capture the regional heterogeneity between structural and functional brain graphs and uses the transformer to fuse the fusion bottleneck and each modality representation. Then, HeBrainGNN [40] considers the heterogeneity between the two cerebral hemispheres. Furthermore, MMGL [11] designs an inter-modal attention matrix for the common representation and directly connects modality representations

Table 1

Comparison with existing multi-modal brain graph learning methods.

Methods	Multi-attribute mining	Pathogenic analysis	Multi-modal relationship mining	Fusion strategy	Model output
SCP-GCN [12]	✗	✗	✗	Graph construction	Common representation
MGNet [9]	✗	✗	✓	Tensor decomposition	Different modality combination
MMGL [11]	✗	✗	✓	Attention mechanism	Common and modality combination
BrainNN [13]	✗	✗	✗	Contrastive learning	Different modality combination
BMCL [23]	✗	✓	✗	Contrastive learning	Different modality combination
Cross-GNN [17]	✗	✓	✓	Self-distillation	Common representation
RH-BrainFS [14]	✗	✗	✓	Fusion bottleneck	Common and modality combination
BrainMM (ours)	✓	✓	✓	Self-expression theory	Common and modality combination

and common representations. For the fusion of imaging and non-imaging data, MM-GTUNets [41] designs modality reward representation learning to construct population graphs. MM-GHiNet [42] combines gene expression data to enhance brain graph node representations. However, they ignore the integration of multi-modal neuroimaging data as studied in this work.

In summary, existing methods cannot consider multi-attribute features for the graph convolution and enhance the quality of brain graph representations. Most of these methods only learn common representations or directly combine different modality representations, ignoring the importance of integrating both representations. Furthermore, they are difficult to achieve a unified framework for capturing the complementary information among multi-modal data and providing the pathogenic analysis. However, the novelty of our method lies in extending multi-attribute features that characterize brain graph properties and designing cross-scale contrastive graph neural networks based on multi-attribute brain graphs to enhance feature representations of each modality. This is conducive to promoting the fusion of multi-modal representations. Besides, our method introduces a self-expression property theory to extract complementary information among different modalities and generate a consensus brain graph for the common feature learning and interpretation analysis.

2.2. Graph contrastive learning

Traditional graph convolutional networks [5] follow the message-passing manner that aggregates node features and its neighbors' features to learn node representations in a semi-supervised learning way. But the reliance on annotated graph data limits models' performance in representation learning. Graph contrastive learning (GCL) as a self-supervised learning method has gained increasing attention and enables graph learning models to produce informative representations from unlabeled graph data. GCL aims to construct contrast samples by data augmentation to enhance graph representation learning under contrast loss optimization. Currently, GCL focuses on the following novelties: graph data augmentation methods [43], graph contrast designs [44], and downstream task analysis [45]. Differently, our method constructs multi-attribute views as contrast samples to enrich brain graph information and designs cross-scale contrast learning based on node-level and graph-level representations to enhance their feature representations.

3. Preliminaries

3.1. Brain graph construction

Here, we introduce the construction process of brain graphs for fMRI and DTI modalities. First, these neuroimaging data are preprocessed according to the pipeline¹, and the main purpose is to eliminate noise information and normalize neuroimaging for all subjects. For fMRI, the preprocessing includes realignment to the first volume, followed by slice timing correction, normalization, spatial smoothness, band-pass filtering, and linear trend removal of the time series. Then, we use the brain atlas, such as automated anatomical labeling (AAL) [46], to compartmentalize different brain regions (ROIs) as nodes of brain graphs. We calculate the mean of all voxels of each ROI to obtain the BOLD signal series of each. We then construct the pairwise connectivity in the BOLD signal series of two ROIs by the Pearson correlation coefficients, i.e., the edge generation. The final undirected weighted graph is generated based on the functional connectivity matrix above. We apply proportional quantization threshold processing to the functional connectivity matrix to obtain the adjacency matrix. For DTI data, the preprocessing includes distorted data removal, head motion correction, brain segmentation, removing irrelevant voxels, voxels associated with brain matter extraction, aligning the diffusion image with the structural image, and reconstructing local diffusion patterns. Finally, a tractographic algorithm, such as deterministic tractography, reconstructs any fiber tract connections between brain regions. We calculate the number of fibers in each ROI parcellated by d cortical/subcortical gray matter regions and build the structure connectivity between ROIs according to the fiber number. So the structure connectivity matrix is an undirected weighted graph of DTI. Similarly, the adjacency matrix is obtained by the threshold processing in the structure connectivity matrix.

3.2. Problem setup

Given a multi-modal neuroimaging dataset, we model them into multi-modal and multi-attribute brain graphs as $M = \{\{G_{1b}, \dots, G_{vb}, \dots, G_{Vb}\}, y_b\}_{b=1}^B$, where G_{vb} represents the brain graph of v th modality in the b th subject. y_b is the class label denoting the healthy or disorder for the subject. Furthermore, $G_{vb} = \{V, E, \mathbf{A}, \mathbf{X}\}$, where $|V| = N$ and $e_{ij} \in E, i, j \in N$ are the node number, i.e., ROIs, and the edge set between ROIs, respectively. $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the adjacency matrix generated by the brain connection matrix that describes the topology structure of brain graphs, where $A_{ij} = 1$ defines that there is an edge link between node v_i and v_j . $\mathbf{X} = \{(\mathbf{X}_1, \mathbf{X}_2) \in \mathbb{R}^{N \times d}\}$ defines two attribute feature matrixes, and d is the feature dimension. Here, the detailed attribute construction methods will be introduced in Section 4.2.

3.3. Self-expression property theory

The self-expression property theory [21] mainly outlines how each data sample can be represented as a linear blend of other samples within the same category. The formulation is as follows:

$$\min_{\mathbf{P}} \|\mathbf{Q}^T - \mathbf{Q}^T \mathbf{P}\|_F^2 + \beta \Theta(\mathbf{P}), \quad (1)$$

¹ <https://braingb.us/preprocessing/>

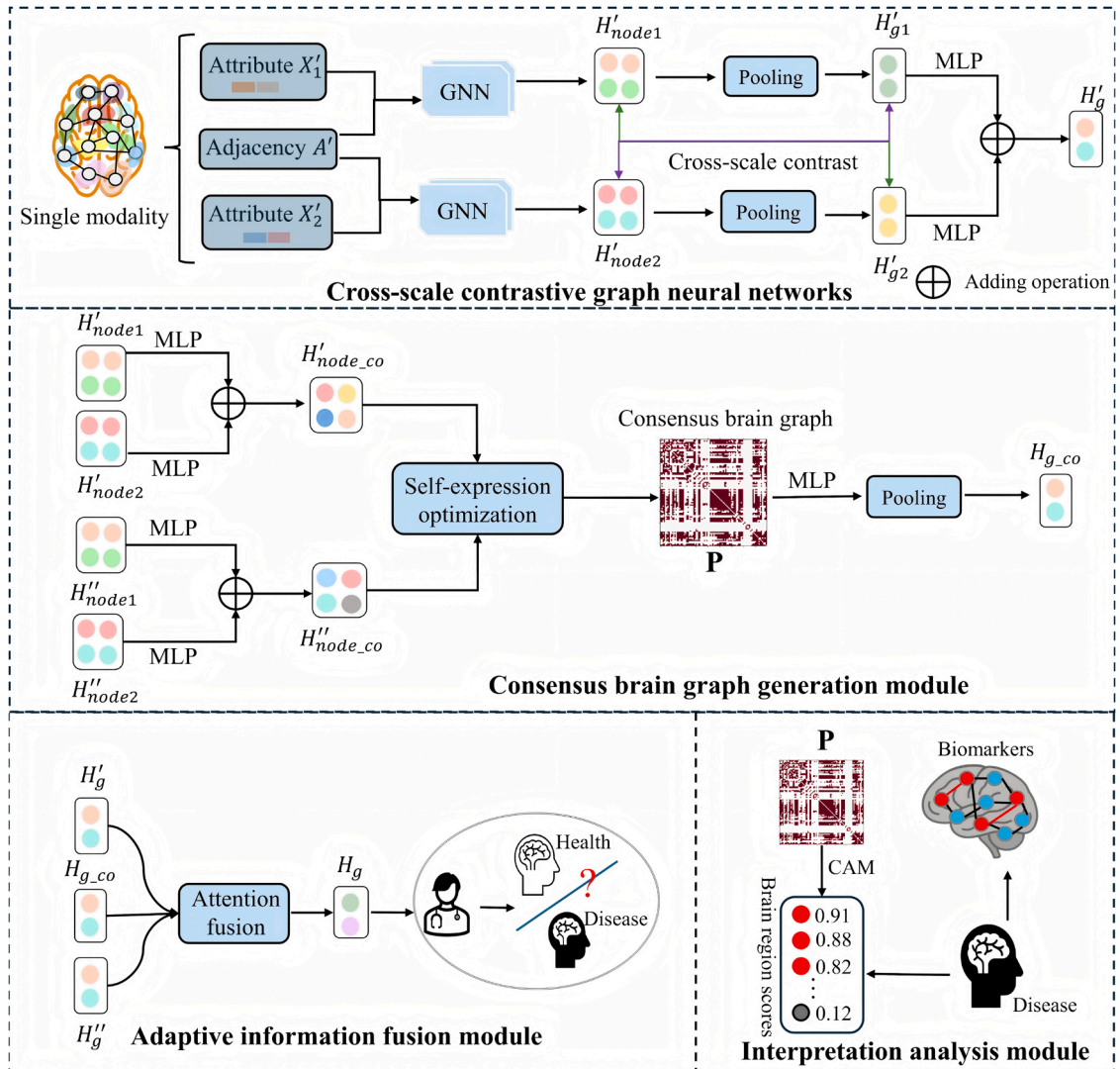


Fig. 1. The whole framework of BrainMM, including cross-scale contrastive graph neural networks, consensus brain graph generation module, adaptive information fusion module, and interpretation analysis module.

where $\mathbf{Q}$ represents data samples and $\mathbf{P}$ is the coefficient matrix that is shared in all samples. $\|\cdot\|_F^2$ is used to measure reconstruction errors. β and $\Theta(\cdot)$ are the trade-off parameter and the regularization function, respectively. Currently, self-expression theory has been employed in many fields, including local subspace learning on temporal data [47] and cross-view subspace clustering [48]. Here, we borrow the key idea of the theory to capture the complementary relationship of multi-modal data while generating the consensus graph for learning the common feature and pathogenic analysis.

4. Method

4.1. Overview

The proposed multi-modal brain graph learning framework BrainMM is shown in Fig. 1. It consists of the following four main parts: (1) **Cross-scale contrastive graph neural networks** inputs two attribute matrices with an adjacency matrix into corresponding GNNs, respectively. Then, it designs cross-scale contrastive learning among node-level and graph-level representations and outputs fused graph-level representations for a single modality. (2) **Consensus brain graph generation module** first fuses node-level representations for each modality and then generates the consensus brain graph under the self-expression optimization to learn the common feature. (3) **Adaptive information fusion module** fuses the common feature and each modality feature by the attention mechanism to learn optimal brain graph representations for disorder prediction. (4) **Interpretation analysis module** uses the consensus brain graph to extract salient ROIs and connections by the CAM method for pathogenic analysis.

4.2. Cross-scale contrastive graph neural networks

Due to the lack of node attribute features in brain graphs, most brain graph learning methods with GNNs [6,7,31] use the adjacency matrix of brain graphs as an attribute feature matrix to conduct the graph convolution for representation learning. However, the single attribute feature is insufficient for the complex brain graph description, especially since this cannot highlight the difference between learned graph representations. Besides, this will influence multi-modal information fusion in the following steps. Inspired by works [2,20], they explore how to construct useful node attribute features for non-attributed graphs. Thus, we extend two attribute feature matrices for brain graphs as two attribute contrast views. Then, we design cross-scale contrastive graph neural networks to enhance multi-level brain graph representations. According to the guidance of the previous work [2], we select two effective construction methods of node attribute features as follows:

- Attribute feature matrix $\mathbf{X}_1$ is defined from the adjacency matrix A . This way mainly considers the topology structure information of brain graphs for message propagation and keeps the similarity between neighboring nodes with the same local structure.
- Attribute feature matrix $\mathbf{X}_2$ is constructed by the eigen decomposition method on the normalized adjacency matrix. We select the top k eigen vectors as the feature information of each node [49] to form the attribute feature matrix, where the value of k is optimized by the grid search. Here, the eigen vector well characterizes the global structural property of each node. It also captures the cluster information to conform to the small-world network property of brain graphs as concluded in [50].

Thus, the attribute features above consider local and global structure information. We jointly fuse the two attribute features to describe the brain graphs of each modality and obtain more extensive brain graph representations than previous methods. Of course, other node attribute feature methods are also discussed in Section 5.4. Then, each attribute feature matrix with the adjacency structure matrix is inputted into GNNs in the single modality view. Here, we use the GCN model [27] for brain graph convolution. The formulation is as follows:

$$\mathbf{H}'_{node1} = f(\phi'_1; \mathbf{X}_1, \mathbf{A}'), \quad (2)$$

$$\mathbf{H}'_{node2} = f(\phi'_2; \mathbf{X}_2, \mathbf{A}'), \quad (3)$$

where $\mathbf{H}'_{node1}$ and $\mathbf{H}'_{node2}$ are node-level representations for the convolution layer output. $f(\phi'_1; \cdot)$ and $f(\phi'_2; \cdot)$ are the parameterized GCN function. Then, we use the readout pooling operation to obtain graph-level representations of them, respectively.

$$\mathbf{H}'_{g1} = g(\psi'_1; \mathbf{H}'_{node1}), \quad (4)$$

$$\mathbf{H}'_{g2} = g(\psi'_2; \mathbf{H}'_{node2}), \quad (5)$$

where $g(\psi'_1; \cdot)$ and $g(\psi'_2; \cdot)$ are the parameterized readout operation like mean-pooling.

To learn better graph-level representations for the single modality, we introduce the cross-scale contrastive learning mechanism based on node-level and graph-level representations in the different attribute views. This simultaneously enhances node-level representations to promote the relationship extraction of multi-modal brain graphs. We first perform the projection head by inputting these node-level representations into a two-layer MLP as $\mathbf{H}'_{node1'}$ and $\mathbf{H}'_{node2'}$. The graph-level representations $\mathbf{H}'_{g1'}$ and $\mathbf{H}'_{g2'}$ are generated by a two-layer MLP as shown in [43]. Then, the contrastive loss $\mathcal{L}_{con1}$ is constructed as follows:

$$\frac{1}{B} \sum_{G_{vb} \in M} \left[\frac{1}{N} \sum_{i=1}^N \left[MI(\mathbf{h}'_{node1'_i}, \mathbf{H}'_{g2'}) + MI(\mathbf{h}'_{node2'_i}, \mathbf{H}'_{g1'}) \right] \right], \quad (6)$$

where $MI(\cdot)$ represents the mutual information between node-level and graph-level representations. Here, we use the deep InfoMAX method [51] to maximize the mutual information between different levels of graph representations from different attribute views. Similarly, for another modality data, we conduct the same operation above and generate corresponding node-level representations as $\mathbf{H}''_{node1'}$ and $\mathbf{H}''_{node2'}$ and graph-level representations as $\mathbf{H}''_{g1'}$ and $\mathbf{H}''_{g2'}$. Thus, through the cross-scale contrastive learning mechanism between node-level and graph-level representations, this not only effectively fuses different attribute features but also improves the quality of node-level and graph-level representations. In addition, this also provides a better basis for consensus graph generation and multi-modal brain graph information fusion.

4.3. Consensus brain graph generation module

Replacing the previous fusion methods that directly fuse different modality representations, we introduce the self-expression property theory to extract the complementary relationship of multi-modal brain graphs while generating a consensus graph to learn the common feature. As shown in the theory of Eq. (1), the coefficient matrix $\mathbf{P}$ describes the relationship between different samples. Similarly, we learn such a shared graph structure pattern between different modalities, i.e., the consensus brain graph, to extract their complementary relationships. Considering the connectivity sparsity of brain graphs and capturing various structural information among ROIs, we hope that the generated consensus graph includes more plentiful brain graph information under the constraint of multi-modal data. Thus, we first design a structure fusion operation to capture different orders of structure proximity in the single modality:

$$f_{order}(G_{vb}) = \mathbf{A} + \mathbf{A}^2 + \dots + \mathbf{A}^T, \quad (7)$$

so for a given brain graph G_{vb} , $\mathbf{A}$ represents the first-order structure proximity with directly connected nodes. The second-order proximity $\mathbf{A}^2$ represents two nodes with the probability of a two-hop random walk. So the higher T -order proximity $\mathbf{A}^T$ shows the probability that a random walk from one node to another with T steps [52].

Next, inspired by Eq. (1), we make the new definition for the consensus graph generation by multi-modal brain graphs:

$$\min_{\mathbf{P}, \kappa^v} \sum_{v=1}^V \kappa^v \left(\|(\mathbf{H}_{node_{co}}^v)^T - (\mathbf{H}_{node_{co}}^v)^T \mathbf{P}\|_F^2 + \beta \|\mathbf{P} - f_{order}(G_{vb})\|_F^2 \right) + \sum_{v=1}^V (\kappa^v)^\tau, \quad (8)$$

where $\mathbf{H}_{node_{co}}^v$ is calculated by the adding operation for $\mathbf{H}_{node1'}^v$ and $\mathbf{H}_{node2'}^v$ in the single modality. V is the whole number of modalities and $f_{order}(G_{vb})$ represents a fusion matrix $\in \mathbb{R}^{N \times N}$. It is worth noting that here we utilize the weight parameter κ^v to distinguish the different contributions of each modality data to the consensus graph. $\tau < 0$ is the smooth parameter. Thus, through node-level representations with cross-scale contrast optimization and multiple high-order structure information guidance, the consensus graph $\mathbf{P}$ not only captures the shared graph structure pattern among different modalities but also keeps the structure properties of brain graphs.

Effectiveness Analysis. Here, we provide a complete explanation of employing self-expression theory for the generation of consensus brain graphs. First, our objective is to capture the complementary information between different modalities and ensure the reliability of the consensus brain graph. As shown in the new Eq. (8) inspired from the self-expression theory, the first term is to minimize the reconstruction error between node representations and their product with the consensus brain graph $\mathbf{P}$, enabling $\mathbf{P}$ to learn the complementary information by integrating all representations of modalities. Besides, a weight parameter κ^v is used to adaptively focus on important modality information. The previous multi-modal fusion methods [10–12] either utilize attention to learn common features or consider different modalities as structural or attributed information for constructing a shared brain graph, so the optimization loss above is a more direct and reasonable way to capture the common features. The second term is a reconstruction error between $\mathbf{P}$ and the fusion matrix $f_{order}(G_{vb})$, and $f_{order}(G_{vb})$ fuses multi-order structure information of brain graphs. This makes $\mathbf{P}$ to preserve the structural property of brain graphs as soon as possible by minimizing the error loss. The third term is the weight parameter regularization. In summary, self-expression theory can better achieve the objective above by using clear loss functions.

4.4. Consensus brain graph optimization

Considering that there are two solved variables to be optimized in Eq. (8), we perform the flexible method to find the solution that updates one variable while fixing the other variables according to the guidance in [19]. The detailed operation is as follows:

- Fix κ^v , Update $\mathbf{P}$: The first-order derivative of Eq. (8) can be set to zero for $\mathbf{P}$, where I is the identity diagonal matrix of the adjacency matrix as:

$$\mathbf{P} = \left(\sum_{v=1}^V \kappa^v (\mathbf{H}_{node_{co}}^v) (\mathbf{H}_{node_{co}}^v)^T + \sum_{v=1}^V \kappa^v \beta I \right)^{-1} \cdot \left(\sum_{v=1}^V \kappa^v \beta f_{order}(G_{vb}) + \sum_{v=1}^V \kappa^v (\mathbf{H}_{node_{co}}^v) (\mathbf{H}_{node_{co}}^v)^T \right). \quad (9)$$

- Fix $\mathbf{P}$, Update κ^v : We first simplify the operation as:

$$\delta^v = \|(\mathbf{H}_{node_{co}}^v)^T - (\mathbf{H}_{node_{co}}^v)^T \mathbf{P}\|_F^2 + \beta \|\mathbf{P} - f_{order}(G_{vb})\|_F^2. \quad (10)$$

Then the objection function is as follows:

$$U(\kappa^v) = \sum_{v=1}^V \kappa^v \delta^v + \sum_{v=1}^V (\kappa^v)^\tau. \quad (11)$$

Thus, we set the first-order derivative of Eq. (11) to zero for κ^v as:

$$\kappa^v = \left(-\frac{\delta^v}{\tau} \right)^{\frac{1}{\tau-1}}. \quad (12)$$

In our experimental optimization process, we conduct certain iterations O to make the objection converge for generating the final consensus graph $\mathbf{P}$.

4.5. Adaptive information fusion module

In these modules above, we learn brain graph feature representations of each modality and the common feature based on the consensus graph. Then, we analyze that brain graph information from different sources has complementary contributions for a better description of one subject's brain, instead of only using the common feature representations in the previous works. Thus, we adopt the attention mechanism to adaptively fuse them. First, we calculate different graph-level representations for these source information, respectively.

$$\mathbf{H}'_g = \mathbf{H}'_{g1'} + \mathbf{H}'_{g2'}, \quad (13)$$

$$\mathbf{H}''_g = \mathbf{H}''_{g1'} + \mathbf{H}''_{g2'}, \quad (14)$$

$$\mathbf{H}_{g_{co}} = MLP(g(\psi'_3; \mathbf{H}_{node_{co}})), \quad (15)$$

where $\mathbf{H}'_g, \mathbf{H}''_g, \mathbf{H}_{g_{co}}$ are graph-level representations from modality 1, modality 2 and the consensus graph, respectively. $\mathbf{H}_{node_{co}}$ is the node-level representations generated by the consensus graph $\mathbf{P}$. Then, we use the attention mechanism to learn important weights for them as:

$$(\pi_1, \pi_2, \pi_3) = att(\mathbf{H}'_g, \mathbf{H}''_g, \mathbf{H}_{g_{co}}), \quad (16)$$

where the function $att(\cdot)$ includes the following steps:

We first transform these graph-level representations by a nonlinear transformation, and then a shared attention vector $\mathbf{q}$ is used to get the attention value ω_1 for representations $\mathbf{H}'_g$ as:

$$\omega_1 = \mathbf{q}^T \cdot \tanh(\mathbf{W} \cdot (\mathbf{H}'_g)^T + \mathbf{b}), \quad (17)$$

where $\mathbf{W}$ and $\mathbf{b}$ are the weight matrix and the bias vector, respectively. Similarly, we get other two attention values ω_2 and ω_3 for representations $\mathbf{H}''_g$ and $\mathbf{H}_{g_{co}}$, respectively. Then, we normalize the attention values by the softmax function to get the importance weight as:

$$\pi_1 = softmax(\omega_1) = \frac{exp(\omega_1)}{exp(\omega_1) + exp(\omega_2) + exp(\omega_3)}, \quad (18)$$

similarly, $\pi_2 = softmax(\omega_2)$ and $\pi_3 = softmax(\omega_3)$. Thus, through the weighted fusion of different source information for each subject, we can get optimized brain graph-level representations $\mathbf{H}_g$ for different subjects, so this enhances the feature representation difference of subjects.

$$\mathbf{H}_g = \pi_1 \cdot \mathbf{H}'_g + \pi_2 \cdot \mathbf{H}''_g + \pi_3 \cdot \mathbf{H}_{g_{co}}. \quad (19)$$

4.6. Objection optimization

To learn effective brain graph-level representations and perform disorder prediction, i.e., brain graph classification, we summarize the following overall objective loss function:

$$\mathcal{L}_{all} = \mathcal{L}_{con1} + \mathcal{L}_{con2} + \mathcal{L}_{cro}, \quad (20)$$

where $\mathcal{L}_{con1}$ and $\mathcal{L}_{con2}$ are the contrast loss from two modalities by Eq. (6). $\mathcal{L}_{cro} = CE(\mathbf{H}_g, y)$ is the cross-entropy loss for graph classification. The whole procedure is shown in Algorithm 1.

Algorithm 1 BrainMM framework.

Input: Brain graphs M , Parameters $T, \beta, \tau, d, l, E, O, B$;
1: Construct $\mathbf{X}_1$ and $\mathbf{X}_2$ for each modality;
2: Initialize $f(\phi'_1), f(\phi'_2), g(\psi'_1), g(\psi'_2), g(\psi'_3)$;
3: **for** $e = 1, 2, \dots, E$ **do**
4: **for** $b = 1, 2, \dots, B$ **do**
5: Sample the batch b of brain graphs;
6: For modality 1, obtain $\mathbf{H}'_{node1}$ and $\mathbf{H}'_{node2}$ via Eq. 2 and Eq. 3, obtain $\mathbf{H}'_{g1}$ and $\mathbf{H}'_{g2}$ via Eq. 4 and Eq. 5;
7: Obtain projection heads $\mathbf{H}'_{node1'}$, $\mathbf{H}'_{node2'}$, $\mathbf{H}'_{g1'}$, and $\mathbf{H}'_{g2'}$ via MLP;
8: For modality 2, obtain $\mathbf{H}''_{node1'}$, $\mathbf{H}''_{node2'}$, $\mathbf{H}''_{g1'}$, and $\mathbf{H}''_{g2'}$;
9: Calculate $\mathcal{L}_{con1}$ and $\mathcal{L}_{con2}$ via Eq. 6;
10: **for** $o = 1, 2, \dots, O$ **do**
11: Compute $f_{order}(G_{vb})$ via Eq. 7 and obtain $\mathbf{H}^o_{node_{co}}$;
12: Optimize Eq. 9;
13: Optimize Eq. 12;
14: **end for**
15: Generate $\mathbf{H}_{g_{co}}$ via Eq. 15, $\mathbf{H}'_g$ and $\mathbf{H}''_g$ via Eq. 13 and Eq. 14;
16: Obtain $\mathbf{H}_g$ via Eq. 19;
17: Calculate $\mathcal{L}_{all}$ via Eq. 20;
18: Back propagation and update parameters;
19: **end for**
20: **end for**
21: Predict the label of $\mathbf{H}_g$ via the classifier;
Output: Consensus brain graph $\mathbf{P}$ and brain graph label $\hat{y}$.

4.7. Interpretation analysis module

As discussed above, multi-modal brain graphs are more conducive to characterizing the brain network condition, so we design a pathogenic analysis method to explore disorder-specific salient brain regions and connections. Specifically, the consensus brain graph

well captures complementary information among different modalities and is suitable as an interpretable brain graph for pathogenic analysis after the whole model is optimized. Inspired by the ENGAGE method [53] that proposes an explainable method to evaluate the node importance for graph data augmentation, here we design a novel unsupervised CAM method to identify important brain region nodes. The original CAM method [54] is an explanation technique for the prediction using visual feature heat maps. And we consider the distribution of node representations of the consensus brain graph to provide prediction explanations as:

$$Score_i = ReLU(\sum_d \omega_d \mathbf{h}_{node_{co}}^{d,i}), \quad (21)$$

where $score_i$ is the importance score of node v_i to evaluate salient brain regions. $\mathbf{h}_{node_{co}}^{d,i}$ is the d th latent dimension of node v_i for the consensus graph. $\omega_d = \left\| \sum_{k \in N_b} readout\left\{ \mathbf{h}_{node_{co}}^i : i \in G_k \right\} \right\|_2 [d]$, where N_b is the set of neighbor graphs that are close to graph-level representations of G_b on the batch graph set.

Thus, through all the importance score values of brain nodes, we can compare the most important brain nodes with healthy and patient brain graphs to analyze which brain regions contribute most to the prediction of brain disorders. Furthermore, we also obtain edge importance scores by computing the mean value of two node importance scores. And then we select the most important edge links in brain graphs to identify salient brain connections. The visualization experiment analysis can be seen in Section 5.9.

4.8. Complexity analysis

The time complexity of our BrainMM mainly includes three parts: First, the whole time complexity of the processing of attribute matrix $\mathbf{X}_2$ is $\mathcal{O}(2k(N + S))$, where k , N , and S are the number of eigen vectors, nodes and edges, respectively. Second, the whole time complexity of two-layer graph convolution, graph pooling, and two-layer MLP operation is $\mathcal{O}((8S + 4N)d + (8 + 16N)d^2)$, where d is the embedding representation dimension. Third, the time complexity of consensus brain graph learning and attention mechanism is $\mathcal{O}((N + 1)d + 2d^2 + 2N^2 + 2TN^3)$, where T is the order number of matrix $\mathbf{A}$. Thus, the total time complexity of BrainMM is $\mathcal{O}(2k(N + S) + (8S + 5N + 1)d + (10 + 16N)d^2 + 2N^2 + 2TN^3)$. Considering that the number of brain graph nodes and embedding dimension are small, so the time complexity is also reasonable. The time cost results are analyzed in Section 5.11.

5. Experiments

In this section, we perform various experiments on three benchmark graph datasets from the public domain to evaluate the effectiveness of BrainMM.

5.1. Experimental settings

5.1.1. Dataset

To practically evaluate the performance of BrainMM, we use three available brain neuroimaging data following previous works [7,32] in the public domains. **Human Immunodeficiency Virus Infection (HIV):** [7] consists of 35 early HIV patients and 35 seronegative controls. Each subject is shown by two modalities: fMRI and DTI. The two modalities are all uniformly parcellated into 90 regions. **Bipolar Disorder (BP):** [7] includes 52 bipolar I individuals and 45 healthy controls across fMRI and DTI modalities. Each brain imaging modality is parcellated into 82 regions. **Parkinson's Progression Markers Initiative (PPMI):** [31] includes three modalities from the Probabilistic Index of Connectivity (PICO), Hough voting (Hough), and FSL² formed by different whole brain tractography algorithms, where each modality 718 subjects, including 569 patients with Parkinson's disease and 149 healthy controls. In addition, each modality's brain graph consists of 84 regions parcellated from T1-weighted structural MRI using the Freesurfer tool. The preprocessing and brain graph construction details of these datasets can be referenced in [2,31], with the corresponding resources also publicly available on BrainGB³, a US-based public benchmark platform for brain graph analysis. Considering the difficulty in collecting multi-modal neuroimaging data, the dataset size is relatively small for HIV and BP and this may affect generalization, while also noting that we used the larger PPMI dataset including 718 subjects to partially mitigate this issue. This research is about fundamental data science research involving graph analysis algorithms from a data mining perspective and does not involve any treatments and interventions for human health or disease.

5.1.2. Baselines

We compare BrainMM with multiple state-of-the-art methods. (1) GNN methods: GCN [27] is a classical graph convolution model that learns node representations by fusing the features of nodes' neighbors. GAT [55] is a variant of GCN [27] to fuse different weights of neighbors' features. GIN [56] learns node representations by the Weisfeiler-Lehman graph isomorphism test. (2) Graph contrastive learning methods: GraphCL [43] designs graph data augmentation methods for graph contrastive learning. InfoGraph [57] enhances graph-level representations by maximizing the mutual information between graph-level representations and the representations of substructures. (3) Single modality brain graph methods: BrainGNN [6] designs a regularized pooling layer to select important node features for graph-level representations. IBGNN [7] learns a globally shared mask across the group to highlight salient ROIs and

² <https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/FSL>

³ <https://braingb.us/>

Table 2

Performance comparison for disorder prediction. * indicates that the performance improvement is statistically significant when BrainMM and the best baseline are compared with a paired t -test level of $p < 0.05$. The best results are bolded, and the runner-up results in baselines are underlined. - means not available.

Methods	HIV				BP				PPMI			
	Accuracy	AUC	Sensitivity	Specificity	Accuracy	AUC	Sensitivity	Specificity	Accuracy	AUC	Sensitivity	Specificity
GCN [27]	0.54±0.15	0.54±0.15	0.57±0.24	0.51±0.07	0.53±0.01	0.53±0.02	0.59±0.28	0.47±0.32	0.63±0.05	0.62±0.07	0.71±0.12	0.55±0.16
GAT [55]	0.64±0.09	0.64±0.09	0.57±0.23	0.71±0.07	0.51±0.01	0.52±0.04	0.35±0.20	0.68±0.17	0.69±0.02	0.66±0.02	0.72±0.13	0.62±0.11
GIN [56]	0.61±0.13	0.61±0.13	0.45±0.33	<u>0.77±0.26</u>	0.58±0.07	0.59±0.05	0.39±0.18	0.80±0.20	0.63±0.05	0.60±0.02	0.66±0.09	0.57±0.06
GraphCL [43]	0.60±0.15	0.59±0.14	0.57±0.18	0.62±0.25	0.59±0.12	0.58±0.12	0.67±0.22	0.51±0.20	0.68±0.05	0.53±0.05	<u>0.79±0.11</u>	0.45±0.17
InfoGraph [57]	0.63±0.05	0.63±0.05	0.60±0.24	0.66±0.25	0.61±0.11	0.60±0.11	0.70±0.26	0.57±0.23	0.72±0.06	0.58±0.05	0.76±0.08	0.57±0.12
BrainGNN [6]	0.61±0.11	0.61±0.11	0.60±0.35	0.63±0.26	0.53±0.10	0.51±0.09	<u>0.72±0.31</u>	0.42±0.24	0.77±0.04	0.62±0.05	0.78±0.21	0.54±0.16
IBGNN [7]	<u>0.64±0.07</u>	0.64±0.07	0.46±0.21	0.72±0.22	0.59±0.11	0.60±0.10	0.41±0.24	0.70±0.20	0.73±0.08	<u>0.68±0.07</u>	0.72±0.11	<u>0.75±0.13</u>
MMGL [11]	0.60±0.10	0.60±0.10	0.51±0.33	0.68±0.22	0.52±0.04	0.53±0.03	0.36±0.21	0.71±0.27	0.66±0.03	0.66±0.02	0.68±0.10	0.64±0.05
MGNet [9]	0.52±0.08	0.52±0.09	0.55±0.14	0.49±0.25	0.64±0.06	<u>0.64±0.06</u>	0.71±0.15	0.57±0.12	0.67±0.04	0.67±0.04	0.71±0.09	0.73±0.07
Cross-GNN [17]	0.62±0.19	<u>0.67±0.15</u>	0.61±0.26	0.70±0.26	0.50±0.15	0.54±0.18	0.53±0.14	0.46±0.24	–	–	–	–
RH-BrainFS [14]	0.58±0.10	0.57±0.08	<u>0.73±0.25</u>	0.41±0.21	0.64±0.07	<u>0.64±0.06</u>	0.57±0.14	0.71±0.12	–	–	–	–
BrainMM_GCN	0.84±0.16	*0.85±0.13	0.85±0.21	0.78±0.19	0.80±0.14	0.83±0.12	0.82±0.11	0.78±0.14	<u>0.76±0.08</u>	0.78±0.12	0.80±0.06	0.74±0.02
BrainMM_GAT	0.81±0.13	0.79±0.17	0.74±0.05	0.82±0.10	0.76±0.14	0.77±0.17	0.74±0.09	<u>0.78±0.04</u>	0.71±0.05	*0.78±0.06	0.79±0.05	0.76±0.11
BrainMM_GIN	0.82±0.05	0.80±0.08	0.85±0.15	0.75±0.11	0.75±0.10	0.79±0.05	0.78±0.06	0.76±0.08	0.75±0.03	0.76±0.09	0.74±0.06	0.76±0.03

connections. (4) Multi-modal brain graph methods: MMGL [11] designs inter-modal attention matrix for the common representation. MGNet [9] uses tensor representation technology to fuse multi-modal brain graphs. Cross-GNN [17] learns a compositional space by constructing a dynamic multi-modal graph considering inter-modal relationships. RH-BrainFS [14] designs a fusion bottleneck to capture the regional heterogeneity between structural and functional brain graphs and uses the transformer to fuse the fusion bottleneck and each modality representation.

5.1.3. Implementations

We adopt GCN [27] with 2 layers and MLP with 2 layers in our framework⁴, where GCN and MLP models are not shared in the same modality. The dimension of brain graph-level representations is 128. The readout pooling operation is sum pooling. According to the performance in the implementation process, T , β , and τ are set to 2, 15, and -1, respectively. The iterations O are set to 20 for the consensus graph generation. We set the batch size as 16 and the training epoch as 100. Adam is used as the optimizer with learning rates 0.01, 0.005, and 0.0001 for HIV, BP, and PPMI, respectively. Specifically, we use 80% of the data for training, 10% for validation, and the remaining 10% for testing. The experiments are repeated five times to report the average results. The experimental results of multi-modal baselines are reported according to their paper settings. For other baselines, we directly fuse graph representations of different modalities by an adding operation and report results with the same setting.

5.2. Disorder prediction performance

From the results in Table 2, we have the following observations: (1) BrainMM consistently outperforms all baselines apart from the Accuracy of PPMI and the Specificity of BP. Besides, the performance improvement is significant on most metrics compared with the results of the baselines. The main reason is that our method first introduces cross-scale contrastive graph neural networks to enhance multi-level brain graph representations. Then, the generated consensus graph captures the complementary relationship of multi-modal information for learning common features. Specifically, adaptive fusion is conducive to learning high-quality brain graph-level representations for disorder prediction. Furthermore, many multi-modal baselines represent better performance on most metrics compared with single modality methods. (2) By selecting different GNNs as the graph encoder model, BrainMM_GCN gets better performance than other encoder models on most datasets. The main reason is that the brain graph structure is very sparse, and the advancement of GIN and GAT for graph representation learning does not work well on brain graphs.

5.3. Ablation study

To verify the effectiveness of the main designs of BrainMM, here we perform ablation studies, and the results are reported in Table 3. “Multi-attribute” represents that cross-scale contrastive graph neural networks are replaced by a GCN model based on the adjacency matrix as a node feature matrix. “Cross-scale contrast” means that there is no cross-scale contrast learning mechanism in contrastive graph neural networks. “Consensus graph” represents that BrainMM only utilizes each modality’s information. “Attention” means that the attention mechanism is removed and the simple adding operation is used. Thus, these different variants of BrainMM, such as BrainMM_1, BrainMM_2, and so on, are denoted by only selecting key designs. Based on the results, we first observe that

⁴ <https://github.com/XuexiongLuoMQ/BrainMM>

Table 3
Ablation study experiments.

Methods	Multi-attribute	Cross-scale contrast	Consensus graph	Attention	HIV		BP		PPMI	
					Accuracy	AUC	Accuracy	AUC	Accuracy	AUC
BrainMM_1		✓	✓	✓	0.79±0.11	0.76±0.05	0.68±0.06	0.71±0.10	0.70±0.08	0.72±0.04
BrainMM_2	✓		✓	✓	0.61±0.12	0.60±0.06	0.67±0.12	0.71±0.15	0.65±0.04	0.70±0.12
BrainMM_3	✓	✓		✓	0.72±0.11	0.76±0.11	0.69±0.13	0.65±0.12	0.71±0.14	0.72±0.08
BrainMM_4	✓	✓	✓		0.77±0.11	0.80±0.12	0.71±0.12	0.80±0.14	0.68±0.01	0.74±0.08
BrainMM	✓	✓	✓	✓	0.84±0.16	0.85±0.13	0.80±0.14	0.83±0.12	0.76±0.08	0.78±0.12

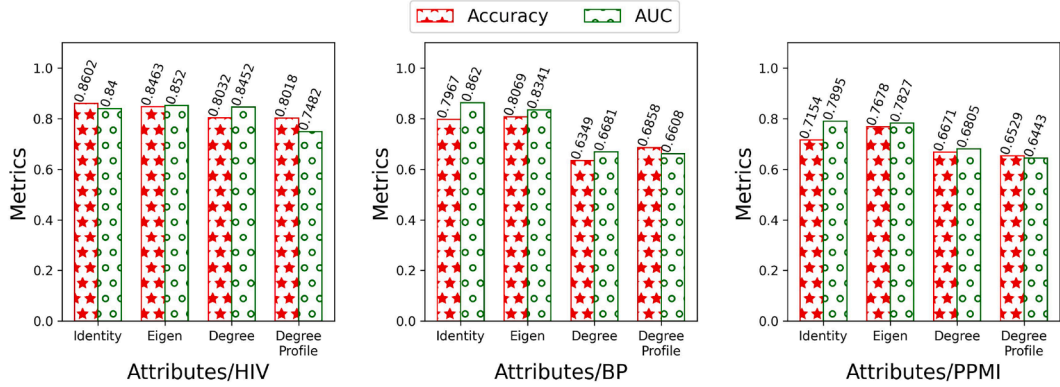


Fig. 2. The exploration of brain node attribute feature methods.

BrainMM still keeps the best performance with all variants on all datasets, which shows that these key designs are very effective for multi-modal brain graph learning. Specifically, there is a large drop in performance for BrainMM_2 and BrainMM_3 on most datasets. This demonstrates that cross-scale contrastive learning can enhance node-level and graph-level representations in multi-attribute settings. The consensus graph generated by the self-expression property theory can well capture the common feature and is indispensable for the final brain graph-level representations.

5.4. Brain node attribute feature analysis

Apart from the eigen method used in the above experiments, for the topology view, we also test other attribute feature methods, including the node identity matrix, the node degree information, and the degree profile information with local statistical measures according to the work analysis [2]. The results are shown in Fig. 2. When the identity is used as another brain node attribute feature, the performance is also competitive. For BP and PPMI, the eigen method achieves better results than other methods. These results show that considering multi-attribute brain graph information is effective for enhancing brain graph representations.

5.5. Information fusion adaptivity

In this section, we perform the density analysis of attention distribution to explore the importance of different source information in the final brain graph representations for all subjects of the three datasets. As shown in Fig. 3, we can see that the attention density of the common feature on HIV and BP is larger than the feature of each modality, which implies that the common feature based on the consensus brain graph is more important than the feature of a single modality. Besides, this conclusion is consistent with the experimental results in Table 3, where BrainMM_3 has a decreasing performance with BrainMM. For the PPMI dataset, the attention density of the single modality feature is larger than the common feature, which shows that brain graph information from the single modality plays a greater role than the consensus brain graph. Similarly, in Table 3, when we remove the consensus graph information from BrainMM, i.e., BrainMM_3, the performance does not have the obvious degradation. In a word, these results demonstrate that BrainMM can adaptively fuse different source information to learn better brain graph representations for disorder prediction.

5.6. Effectiveness analysis of consensus brain graph

Being different from the previous multi-modal brain graph methods that directly integrate different modalities into common brain graph representations, our method aims to learn a consensus brain graph to capture the common feature information. Here, we evaluate the effectiveness of the consensus brain graph in the following two aspects:

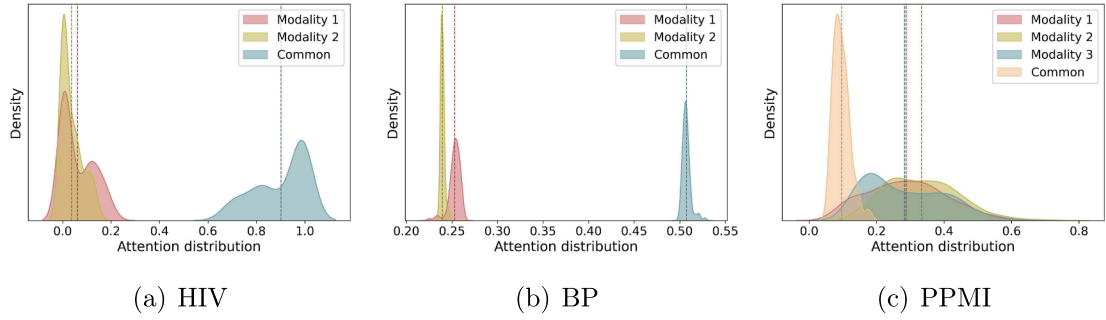


Fig. 3. Analysis of attention distribution to evaluate the importance of different source information. Average attention values of different modalities are marked with the corresponding colors of dashed vertical lines.

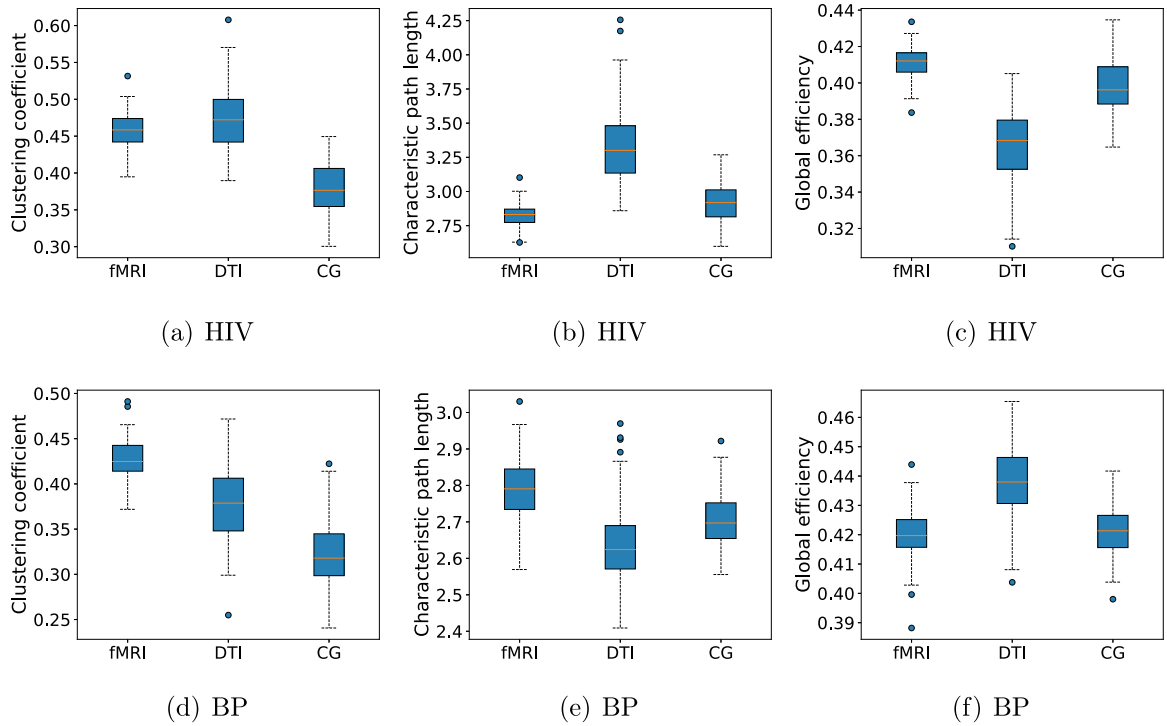


Fig. 4. The evaluation of the generated consensus graph is based on the characteristics of brain networks in terms of clustering coefficient, characteristic path length, and global efficiency, where “CG” denotes the generated consensus brain graph.

5.6.1. Qualitative evaluation

To verify that the consensus brain graph conforms to the basic characteristics of the brain network, we compare three vital measures of brain networks with original brain graphs based on fMRI and DTI. The qualitative results are shown in Fig. 4. We can observe that these measures of the consensus brain graph all keep small errors with other brain graphs, and the results are reasonable according to related brain network analysis [58]. This reflects that the generated consensus brain graph is reliable and can be used as the interpretable brain graph for pathogenic analysis.

5.6.2. Quantitative evaluation

To verify the superiority of our multi-modal learning strategy, we use the consensus brain graph to obtain brain graph representations for disorder prediction, and the results are reported in Table 4. From the contrast results with multi-modal baselines, we can see that though we only use the consensus brain graph instead of the fusion of different modality information, this consensus graph also achieves competitive performance compared with advanced multi-modal fusion baselines. These results are consistent with Fig. 3 that the consensus brain graph plays an important role in multi-modal information fusion. Thus, these results demonstrate that the proposed self-expression theory is beneficial for the common feature mining.

Table 4

The disorder prediction performance based on the consensus brain graph.

Methods	HIV		BP	
	Accuracy	AUC	Accuracy	AUC
MMGL [11]	0.60 $\pm$ 0.10	0.60 $\pm$ 0.10	0.52 $\pm$ 0.04	0.53 $\pm$ 0.03
MGNet [9]	0.52 $\pm$ 0.08	0.52 $\pm$ 0.09	0.64 $\pm$ 0.06	0.64 $\pm$ 0.06
Cross-GNN [17]	0.62 $\pm$ 0.19	0.67 $\pm$ 0.15	0.50 $\pm$ 0.15	0.54 $\pm$ 0.18
RH-BrainFS [14]	0.58 $\pm$ 0.10	0.57 $\pm$ 0.08	0.64 $\pm$ 0.07	0.64 $\pm$ 0.06
Consensus Graph	0.82$\pm$0.09	0.80$\pm$0.05	0.79$\pm$0.09	0.74$\pm$0.06

Table 5

Performance contrast of cross-scale contrastive graph neural networks with single modality baselines.

Methods	HIV-fMRI		HIV-DTI		BP-fMRI		BP-DTI	
	Accuracy	AUC	Accuracy	AUC	Accuracy	AUC	Accuracy	AUC
GCN [27]	0.60 $\pm$ 0.16	0.62 $\pm$ 0.11	0.61 $\pm$ 0.17	0.62 $\pm$ 0.10	0.60 $\pm$ 0.05	0.60 $\pm$ 0.08	0.52 $\pm$ 0.06	0.51 $\pm$ 0.05
DGCNN [59]	0.64 $\pm$ 0.13	0.64 $\pm$ 0.08	0.67 $\pm$ 0.06	0.67 $\pm$ 0.10	0.61 $\pm$ 0.05	0.62 $\pm$ 0.09	0.60 $\pm$ 0.06	0.62 $\pm$ 0.07
BrainGNN [6]	0.66 $\pm$ 0.11	0.63 $\pm$ 0.07	0.64 $\pm$ 0.04	0.61 $\pm$ 0.06	0.63 $\pm$ 0.08	0.61 $\pm$ 0.05	0.59 $\pm$ 0.11	0.56 $\pm$ 0.06
IBGNN [7]	0.68 $\pm$ 0.14	0.74 $\pm$ 0.07	0.66 $\pm$ 0.10	0.66 $\pm$ 0.08	0.61 $\pm$ 0.06	0.61 $\pm$ 0.07	0.59 $\pm$ 0.13	0.60 $\pm$ 0.08
Ours	0.80$\pm$0.13	0.75$\pm$0.06	0.73$\pm$0.09	0.76$\pm$0.09	0.74$\pm$0.05	0.76$\pm$0.09	0.78$\pm$0.11	0.82$\pm$0.10

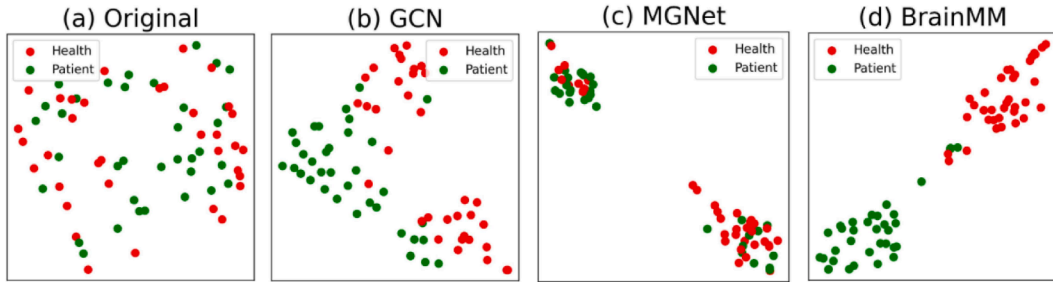


Fig. 5. The visualization of brain graph-level representations on HIV dataset.

5.7. Effectiveness analysis of cross-scale contrastive GNNs

In this section, we mainly test the performance of the proposed cross-scale contrastive graph neural networks, so we conduct analysis experiments on the single modality data. Specifically, we compare single modality baselines, where DGCNN [59] is a graph pooling method. As shown in Table 5, our method achieves better prediction performance than other baselines, especially the improvement is obvious on HIV-DTI, BP-fMRI, and BP-DTI. This demonstrates that our design of introducing multi-attribute information and cross-scale contrast learning is conducive to characterizing brain graph features and learning better brain graph representations.

5.8. Representation visualization

To make a better contrast between brain graph-level representations learned by baselines (GCN and MGNet) and our method, we visualize the representation results. This also includes the original brain graphs for the HIV dataset in Fig. 5. We can see that the boundary of BrainMM between healthy and patient brain graphs is very obvious, and there is minimal overlap among different labels of brain graphs compared to other baselines, which also verifies that our method can learn discriminative brain graph representations for disorder prediction. This benefits from the clever design of cross-scale contrastive learning. It not only enhances graph-level representations of single modality for the following multi-modal fusion, but also ensures the quality of node-level representations for the consensus brain graph generation.

5.9. Case study for pathogenic analysis

5.9.1. Salient brain regions

Through the generated consensus brain graph among multi-modal data, we propose an unsupervised CAM method to get the importance scores of all brain regions as shown in the a and b parts of Fig. 6, where the detailed names of brain regions for BP can be

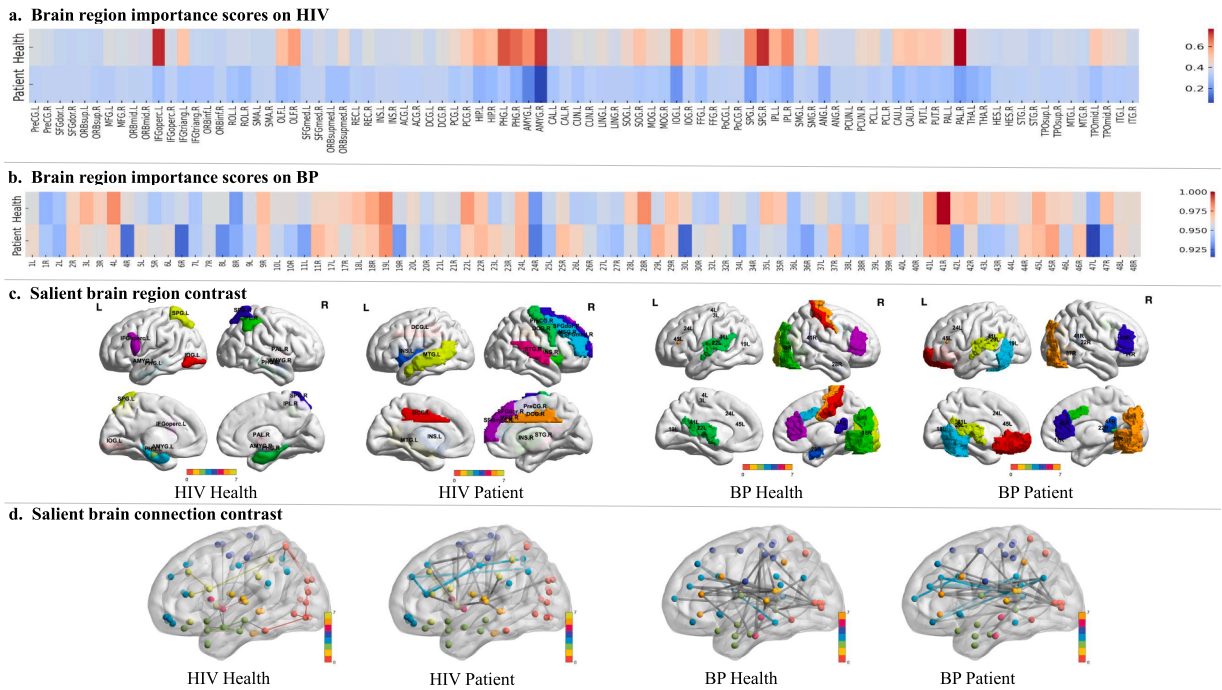


Fig. 6. The case study of pathogenic analysis. **a** and **b** parts are heat maps to represent the importance scores of all brain regions of given healthy and patient subjects on HIV and BP datasets, and “L” and “R” denote the left brain and right brain of humans, respectively. **c** part shows salient brain regions between healthy and patient subjects by ranking the top 10 important brain regions based on their importance scores. **d** part shows salient brain connections between healthy and patient subjects by ranking the top 50 important brain connections based on the importance scores of brain connections, where balls of different colors are the divisions of the nervous system.

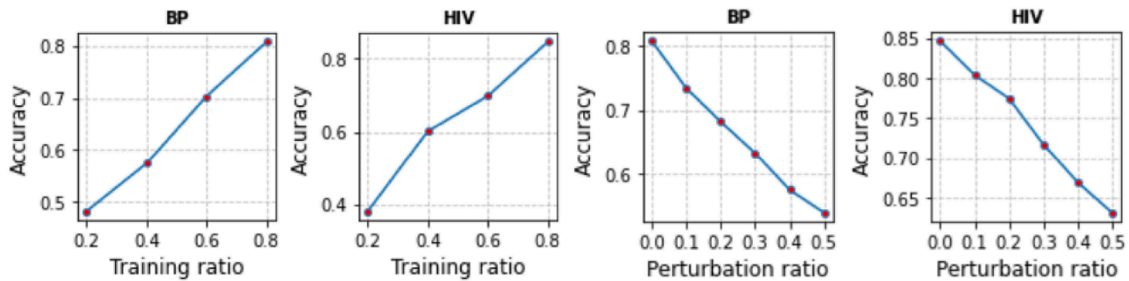


Fig. 7. The influence of different training ratios and perturbation ratios.

seen in the site⁵. We can observe that for the HIV dataset, the gap in importance scores between healthy and patient subjects is large, and the difference in important brain regions is very obvious. However, for the BP dataset, the importance scores between healthy and patient subjects are very close, and the difference in important brain regions is not very obvious. To further analyze salient brain regions between healthy and patient subjects for pathogenic region exploration, we rank the top 10 important brain regions based on their importance scores. The visualization results are shown in the **c** part of Fig. 6. We can see that salient brain regions mainly manifest the frontal gyrus, medial paracingulate gyrus, anterior central gyrus, and temporal gyrus for HIV disorder. Besides, we find that these related medical researches [60] also show that significant changes are occurring mainly in the paracingulate gyri and frontal gyrus. For BP disorder, the associative visual cortex, superior temporal gyrus, fusiform gyrus, anterior cingulate cortex, and auditory cortex are salient regions, which complies with the findings that there is significant hyperconnectivity in the anterior cingulate cortex [61] and BP patients have visual processing disorder [62].

5.9.2. Salient brain connections

To analyze the pathogenic mechanism from brain connections, we compute the connection importance scores based on the importance scores of brain regions. We select the top 50 important connections based on their edge importance scores, and the contrast results between healthy and patient subjects are shown in the **d** part of Fig. 6. For HIV disorder, the connections of Visual Network

⁵ https://en.wikipedia.org/wiki/Brodmann_area

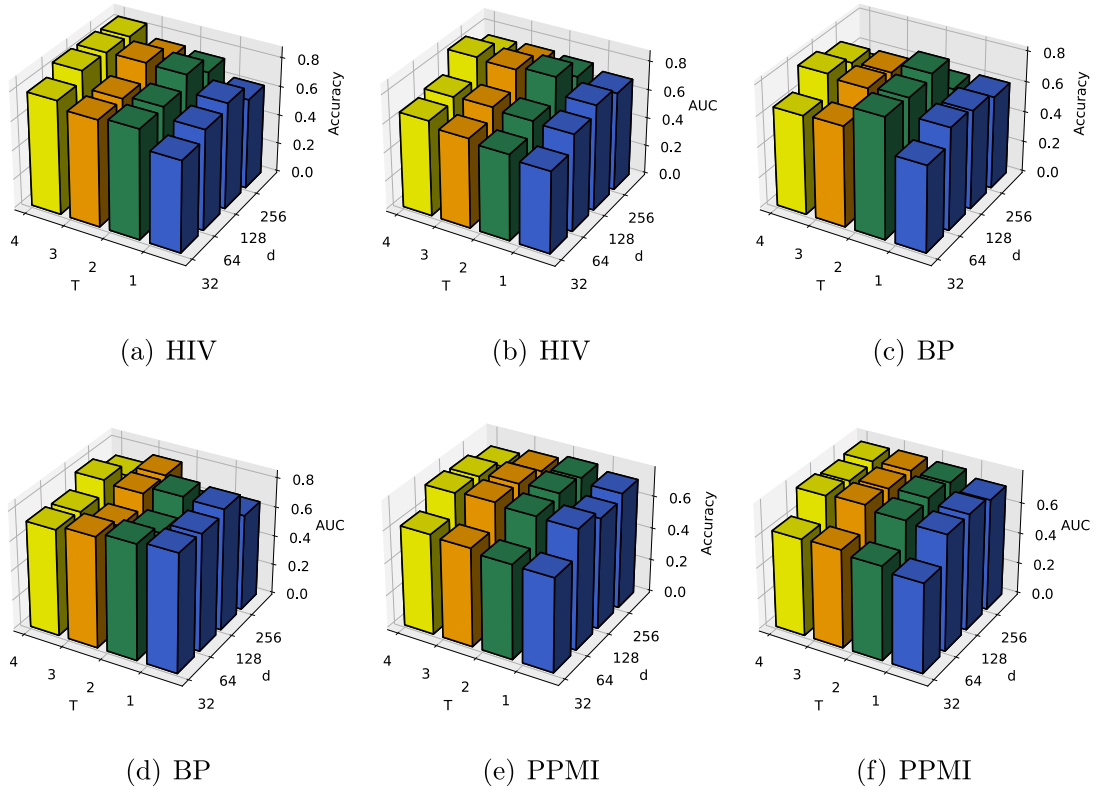


Fig. 8. Parameter analysis about the structure order T in the consensus graph generation and brain graph-level representation dimension d .

(VN), Bilateral Limbic Network (BLN), and Cognitive Control Network (CCN) are decreased, and the connections of Default Mode Network (DMN) are increased. This conclusion is in line with the scientific research results that there are multiple affected connections in DMN [63] and HIV patients show the connection change in VN for visual processing disorder [64]. For BP disorder, we can see that there are increased connections in DMN and decreased connections in BLN. The BP disorder reports [65] also verify the conclusion that the connection within the BLN system is abnormal in clinical analysis.

5.10. Robustness analysis

To evaluate the robustness of our method against random perturbations in the connectivity matrices according to the modality, we randomly add or drop a certain edge ratio in the A of each modality for the datasets of BP and HIV. As shown in Fig. 7, we can see that although the accuracy values start to decrease with the increasing perturbation ratios, our method does not perform poorly in such large perturbations, especially representing competitive performance in small perturbations. Furthermore, we also analyze the reason for the performance decrease is due to the small size of brain graphs and sparse connections. Random perturbations easily produce more patient brain graphs that cannot be recognized.

In addition, we also test the performance of different training ratios of samples for BP and HIV reported in Fig. 7. We observe that more training samples will lead to better performance. Furthermore, due to the whole training sample being small, decreasing the training ratios from 0.8 to 0.6 also does not show poor performance.

5.11. Time complexity

The section mainly measures the efficiency of BrainMM, so we report the average training time of one epoch with multi-modal baselines as shown in Table 6. From the time comparison results, although BrainMM includes multi-attribute fusion for a single modality brain graph, the time costs are not extremely high compared with baselines on three datasets. Besides, the time cost depends on the size of the brain graphs, but the number of brain region nodes is small. Thus, our method is acceptable in real applications.

5.12. Parameter sensitivity

We analyze the change of structure order T in the consensus brain graph generation for disorder prediction performance. Besides, the dimension size d of learned brain graph-level representations is also discussed. All experimental results are shown in Fig. 8. As

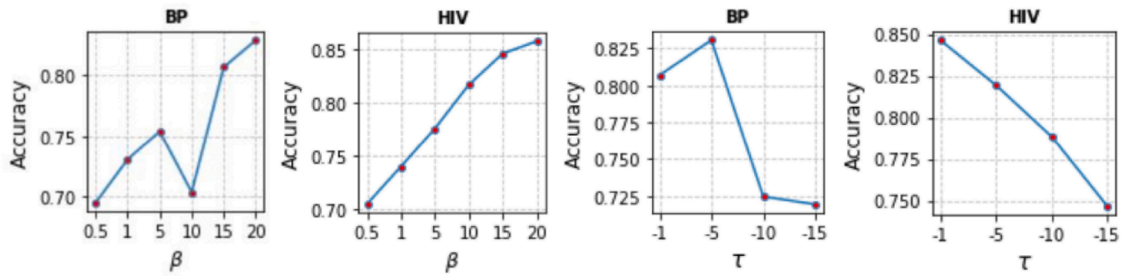


Fig. 9. The analysis in terms of parameters β and τ .

Table 6

Statistics (in second) of the average training time of one epoch for baselines.

Dataset	MMGL	MGNet	Cross-GNN	RH-BrainFS	BrainMM
HIV	0.356	0.315	0.107	0.226	1.388
BP	0.744	0.512	0.137	0.208	1.839
PPMI	13.258	8.148	–	–	42.024

can be observed, when the number of structure orders is larger, the performance of disorder prediction is better, especially when the order T is set to 2 to get better performance on most datasets. The results reflect that, due to the sparsity of brain graph structure, various orders of graph structure from different modalities benefit the consensus graph generation and then improve the quality of brain graph-level representations. For the parameter d , higher dimensions of graph representations get better performance.

In addition, we also explore the influence of parameters β and τ on BP and HIV reported in Fig. 9. As the β value increases, the accuracy starts to increase, and the τ shows a similar trend. The larger β is conducive to enhancing the quality of the consensus brain graph as shown in the experimental results. Thus, we set these parameters according to their represented performance.

6. Conclusion

This work proposes a multi-modal brain graph learning framework with multi-attribute information for disorder analysis. We design cross-scale contrastive graph neural networks to enhance multi-level representation learning and introduce the self-expression property theory to generate the consensus brain graph for complementary relationship mining of multi-modal brain graphs. Besides, different modality features and the common feature are adaptively fused by the attention mechanism to learn optimal graph-level representations for disorder prediction. The consensus brain graph is used to explore disorder-specific salient brain regions and connections. Extensive experiments verify the superiority of BrainMM for disorder prediction. In addition, the results of pathogenic analysis also show that our method is promising in developing an effective disorder analysis technique.

However, there are some limitations. We only explore qualitative interpretations to evaluate salient brain regions and connections related to brain disorders, and a medical cross-validation will be conducive to enhancing the reliability of our method. In future work, we will try more medical analysis with neuroscience experts to promote the real application of this method.

CRedit authorship contribution statement

Xuexiong Luo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation; **Jia Wu:** Writing – review & editing, Supervision, Project administration; **Sheng Zhang:** Methodology; **Jian Yang:** Writing – review & editing, Supervision; **Amin Beheshti:** Writing – review & editing, Resources; **Bo Du:** Writing – review & editing; **Shan Xue:** Writing – review & editing; **Quan Z. Sheng:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link of my code in the manuscript.

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